Building and Deploying AirHeroes

If you would like to re-build the project or make any changes to the project you must first build it before deploying it, however if you only want to deploy an existing build then skip to the deploying section.

Building

Setting up the environment

You will need to install a few different programs in order to build AirHeroes, the most important being Unity. Unity is the tool that we used to build the game, in order to install unity you must first install Unity Hub, Unity Hub can be installed here:

https://unity.com/products/unity-personal

In the top right corner of this website there is a button labeled “Download Unity”, press that, from there you will need to create a Unity account and sign in. You will then be prompted to download Unity Hub, make sure you install it for your given operating system. Once installed you will be prompted to sign into Unity Hub with the Unity account you just created, do so.

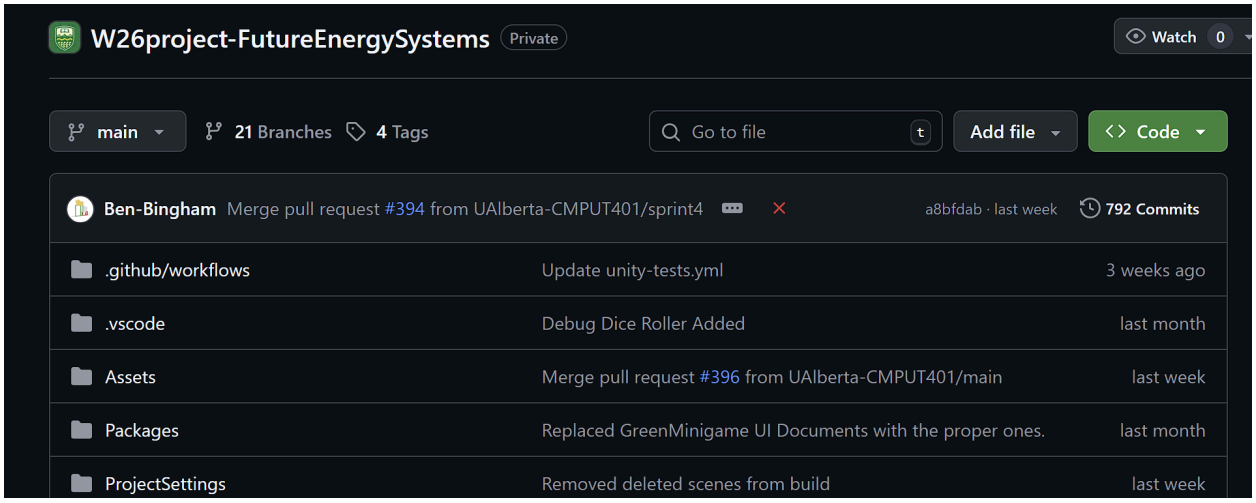
Downloading the code base

The code base is the code used to create the game, using the code base with unity will allow you to build the final game. Begin by navigating here:

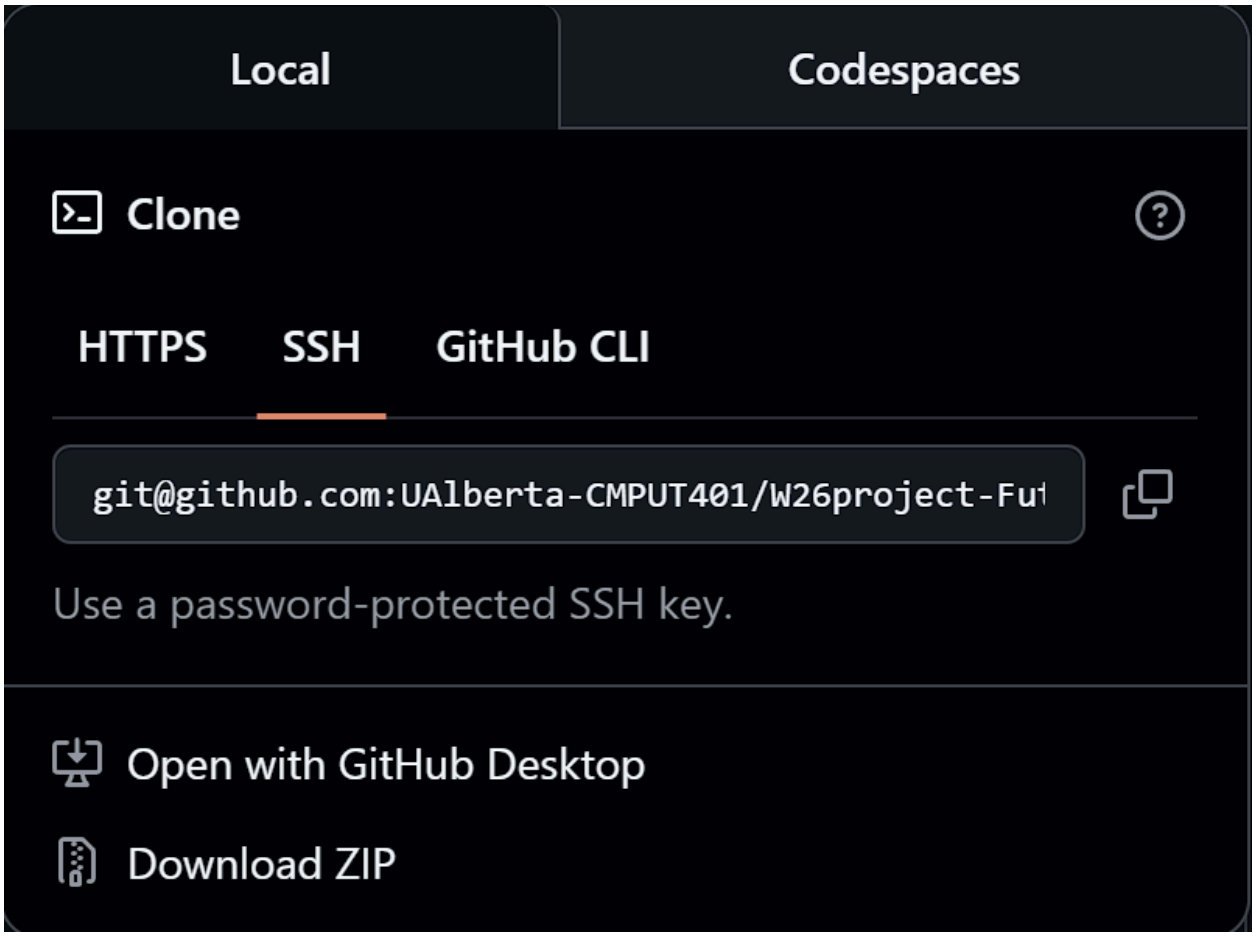
https://github.com/UAlberta-CMPUT401/W26project-FutureEnergySystems

This is the github repository for the code, if you are already familiar with git and github then clone this repository into a local folder on your computer.

If you are not familiar with git and have no plans on editing the code then you can download it as a zip file by first clicking the green “<> Code” button visible in the top right corner of this image:



Next, press the “Download zip” button visible here:



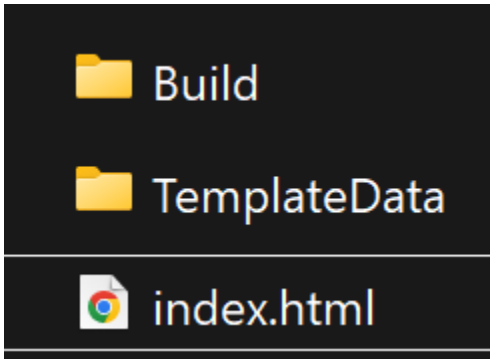
Once downloaded, extract the zip file

Building

Navigate to Unity Hub and under the projects tab select “Add”, then “Add project from disk” in the drop down. When Unity asks for what file to open, navigate to the project folder and select open. Unity hub will now prompt you to download the Unity Editor, be sure to download version “6000.3.4f1”.

Once the editor loads (this may take a few minutes) in the top bar press “File” then “Build and run”, this will ask you where you want the build files to go, create a folder in the root of the project called “build” and select it as the destination (Building may take anywhere from 5 to 30 minutes plus).

Once the build completes it will launch the game in your local web browser, this is helpful for testing that everything works as expected before publishing. In order to upload the build to itch.io we need to compress all of the needed files into one zip file. Navigate to your local installation of the project code and go into the “build” folder you created, you should see these files:



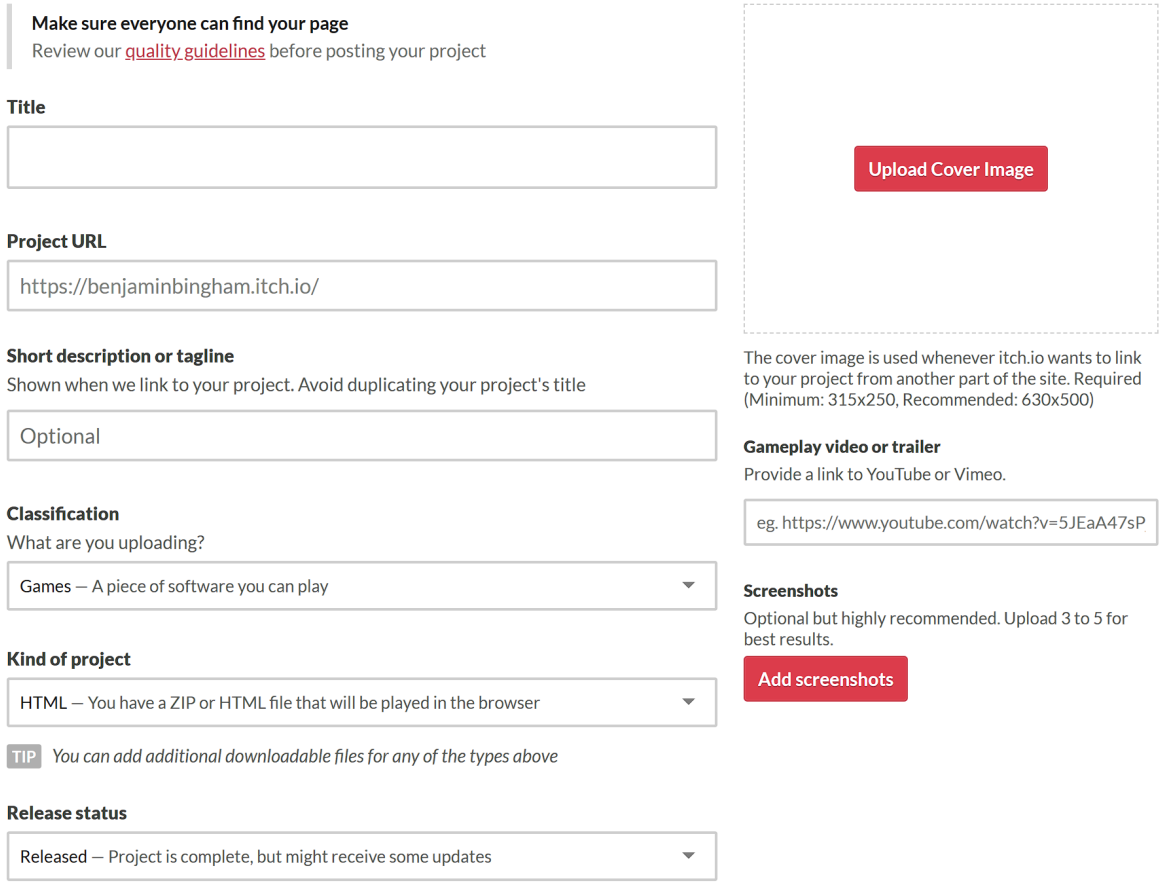
Compress all of them into one file titled “build.zip”, you will need this file later during the upload step.

Deploying

Navigate to:

https://itch.io/

Create an account (Please choose your account name wisely, the URL that the game will be accessible at will be “https://<Account Name>.itch.io/<Game Title>” where <Account Name> is the name of the account and <Game Title> is the title of the game), and then navigate to your dashboard (Accessible in the top bar) and press “Create new project”. This page has many fields, most have information about the game visible to players that can be filled out in any way while other fields must be filled out in a particular manner, lets go over all of the fields one by one (Note, all of these fields can be edited in the future):



Fields that must be set a certain way in order for the game to be playable are underlined.

Title: The title of the game, this will be visible to all players and will be appended to the end of the Project URL

Project URL: This is the URL that players will navigate to in order to play the game

Short description or tagline: This is shown when itch.io links to the game along with the game title

Classification: This must be set to “Games”

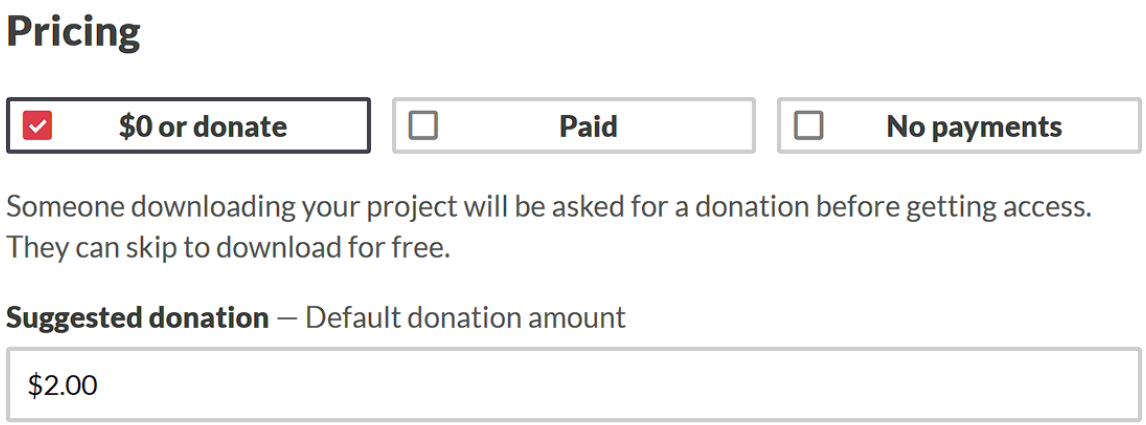
Kind of project: This must be set to “HTML”

Release Status: The release status of the project should be set to “Released” as that is the state of the project, but it is not required to be set that way. If you would like you can set it to another option

Cover Image: This is the main image used to represent the game and will be shown on itch.io

Gameplay video or trailer: This can be a link to a youtube or vimeo video and will be shown on the home page of the game before it is played, this can be used to show off more information about the game

Screenshots: Another place to store information about your game, these screenshots will be visible to players before the game is played.



Pricing: It does not look like itch.io supports payments for games that do not need to be downloaded, regardless if you want the game to be free then set this to “No Payments”.

Uploads

Upload a `ZIP` file containing your game. There must be an `index.html` file in the `ZIP`. Or upload a `.html` file that contains your entire game. Learn more →

Any additional files you upload will be made available for download. You can apply a minimum price to the project after uploading additional downloadable files.

Upload files or Choose from Dropbox Add External file ?

File size limit: 1 GB. Contact us if you need more space

TIP *Use **butler** to upload files: it only uploads what's changed, generates patches for the itch.io app, and you can automate it. Get started!*

Embed options

How should your project be run in your page?

Embed in page | Auto-detect size (Unity HTML only)

Uploads: Press the upload button and then navigate to the “build.zip” file that you built earlier and upload the file.

Embed Options: As shown in the above screenshot, the first option should be set to “Embed in page” and the second to “Auto-detect size (Unity-HTML only)”

Frame options

- ☐ Mobile friendly — Your project can run on mobile phones (smaller resolution and touch support)
- ☐ Automatically start on page load — Not recommended for Unity games, since they can lag the browser when loading
- ☐ Fullscreen button — Add a button to the bottom right corner of your embed to make it fullscreen
- ☐ Enable scrollbars — Enable scrollbars in the iframe that contains your project
- ☐ `SharedArrayBuffer` support — **(Experimental)** This may break parts of the page or your project. Only enable if you know you need it. Learn more

Frame Options: Leave these all blank

Description — This will make up the content of your game page.

Genre

Select the category that best describes your game. You can pick additional genres with tags below

No genre

Tags — Tips for choosing tags

Any other keywords someone might search to find your game. Max of 10.

Avoid using the genre or platforms provided above.

Click to view options, type to filter or enter custom tag

Description: This will be visible before and while playing the game, here you can describe the basics of how to play the game and/or a brief description of the game so that players know what to expect before they hit play.

Genre: The genre of the game, select any option from the dropdown list

Tags: These tags are used when people search for the game on itch.io, in order to help people find the game as easily as possible select tags that describe the game well.

AI generation disclosure NEW — Learn more →

Please disclose if this project contains content produced by generative AI tools such as LLMs, ChatGPT, Midjourney, Stable Diffusion, etc., even if you hand-edited it.

- ○ Yes — This project contains the output of Generative AI
- ○ No — This project does not contain the output of Generative AI

App store links

If your project is available on any other stores we'll link to it.

+ Steam | + Apple App Store | + Google Play | + Amazon App Store | + Windows Store

Custom noun

You can change how itch.io refers to your project by providing a custom noun.
Leave blank to default to: 'game'.

Optional

Community

Build a community for your project by letting people post to your page.

- ○ Disabled
- ◉ Comments — Add a nested comment thread to the bottom of the project page
- ○ Discussion board — Add a dedicated community page with categories, threads, replies & more

Visibility & access

Use Draft to review your page before making it public. Learn more about access modes

- ◉ Draft — Only those who can edit the project can view the page
- ○ Restricted — Only owners & authorized people can view the page
- ○ Public — Anyone can view the page

AI generation disclosure: The blue card images are AI generated and so this must be set to "Yes"

App store links: Links to other places where people can find the game, this can be left blank

Custom noun: This can be left blank, if its changed the game will not be referred to as a game but instead as a <Custom Noun> where <Custom Noun> is the noun you enter here

Community: Enable or disable comments or a discussion board on the page

Visibility and Access: This must be changed to either “Restricted” or to “Public”, if set to “Restricted” then anyone with the link will only be able to access the game with the correct password that you set. If this is set to “Public” then anyone with the link will be able to play the game.

After all of the fields are configured press “Save” at the bottom of the page, the game is now live and accessible from the project URL that is listed in the project URL field, or by pressing the “View Page” button next to the “Save” button you can jump straight to the game.